Evaluating EEG Foundation Representations: Cross-Subject Decoding, Montage Sensitivity, and Latent Structure

Abstract

EEG foundation models seek transferable representations of heterogeneous multivariate time series, but downstream accuracy alone does not establish that their latent features capture task-relevant physiology. Random masking, reconstruction targets, and the coupling of spatial and temporal learning may preserve neural information alongside subject and acquisition effects. This study examines eight pretrained models and a compact supervised EEGNet baseline. Data preparation covers 14 EEG corpora spanning clinical, motor-imagery, affective, sleep, and speech-related tasks; the reported comparisons concern four-class BCI Competition IV-2a, with additional REVE similarity diagnostics on FACED and PhysioNet motor imagery. Average-pooled frozen features yield 26.80–33.31% balanced accuracy with a linear SVM, against a 25% chance reference. Under a separate subject-disjoint protocol, a common token-preserving readout reaches 65.21% with REVE, while EEGNet, with approximately 3,000 parameters, reaches 76.77%. Native channels favor CBraMod and EEGMamba, whereas mapping favors EEGNet. REVE visualizations exhibit subject-associated organization, weak class contrast, and lower held-out energy capture by leading components; these observations do not identify artifacts or prove that task information is absent. Different subject splits, readout capacities, and input adaptation prevent attributing score differences solely to pooling. The findings motivate stronger physiological validation, subject-level replication, and controls for high-dimensional representation metrics. Waveform reconstruction is an imperfect surrogate for physiological relevance, but latent alignment alone is not sufficient either. Non-contrastive joint-embedding methods such as LeJEPA, potentially combined with reconstruction, merit controlled comparison using augmentations that vary nuisance components while preserving neural phenomena. Progress should be judged by interpretable and robust cross-subject transfer, rather than model scale or a single decoding score.

1 Introduction

Electroencephalography (EEG) is a multivariate time series in which temporal dynamics and relationships between electrodes jointly carry information. A useful general-purpose representation should retain task-relevant activity across participants, sessions, electrode layouts, and recording contexts. Large-scale self-supervised pretraining aims to reduce dependence on labels, but transfer requires more than reconstructing observations or obtaining a high score on one dataset. The central question is which aspects of the signal are preserved and remain useful under changes in the data distribution.

1.1 Foundation Models and Spatiotemporal Learning

The comparison includes eight models: CBraMod, EEGMamba, CSBrain, LuMamba, LUNA, LaBraM, EEGPT, and REVE. These models represent different approaches to combining spatial and temporal information, rather than a single homogeneous class of encoders. EEGNet provides a compact supervised comparator [10]. Its task-specific training also distinguishes the value of pretrained features from the performance achievable by learning directly from labels.

LaBraM models channel–time patch sequences with a Transformer and predicts discrete codes from a tokenizer trained through neural spectrum prediction. Its targets therefore differ from direct waveform samples [8]. CBraMod instead separates spatial and temporal interactions into parallel criss-cross attention mechanisms, with positional encoding adapted to the two axes, and reconstructs masked signal patches [17]. CSBrain extends multiscale modeling through temporal windows, anatomical regions, and structured sparse attention linking these scales [20].

EEGPT combines local spatiotemporal encoding with representation alignment to a momentum encoder and a masked reconstruction objective. It consequently already couples signal-level supervision with latent targets [16]. REVE uses Transformer processing with positional information that combines three-dimensional electrode coordinates and temporal position. Its masked autoencoding uses contiguous spatiotemporal blocks, rather than only independently missing patches [7].

EEGMamba uses a bidirectional state-space backbone to model dependencies among EEG patches. Its pretraining objective is masked reconstruction [18]. LUNA first unifies variable electrode sets through learned-query cross-attention, then applies temporal Transformer processing in a fixed latent space [6]. LuMamba uses this channel-unification principle with bidirectional Mamba processing and studies reconstruction, LeJEPA, and their combination [2]. Thus, joint patch modeling, separate axial attention, multiscale

regional processing, and spatial compression before temporal modeling impose different assumptions about how EEG dependencies should be learned.

1.2 What Should Self-Supervision Preserve?

Random masking is a common pretraining mechanism: selected patches are hidden and inferred from the observed context. The prediction target can be a waveform, a discrete code, or a learned representation. Mask location, extent, and spatial organization determine which dependencies are useful for this task. Independent masks can leave nearby temporal samples or correlated channels visible; structured masks can demand broader context. Neither choice guarantees that successful prediction requires the physiological phenomenon relevant to a downstream task.

Waveform reconstruction is particularly difficult to interpret as a physiological learning criterion. EEG measurements combine multiple sources, and task-relevant rhythms or transients need not dominate sample-wise error. For a squared reconstruction loss, larger residual amplitudes receive more weight, regardless of their neural or artifactual origin. Predictable background activity or contamination can therefore be useful for minimizing the objective. Conversely, reproducing the exact phase of a waveform may not be necessary for a task defined by a spectral or spatial pattern. These are reasons to question reconstruction loss as a surrogate for physiological relevance, not to dismiss temporal signals as uninformative.

Latent-space objectives change the target but do not automatically solve this problem. LeJEPA aligns related views and uses distributional regularization to oppose collapse without explicit negative pairs [1]. Contrastive methods such as TS2Vec separately demonstrate the potential of view-based learning for multivariate time series [19]. In EEG, LuMamba reports benefits from combining LeJEPA with reconstruction [2]; this argues for comparing complementary objectives rather than presuming that latent alignment universally supersedes signal reconstruction.

Across these approaches, agreement can preserve subject identity or acquisition effects if both views share them. Spatial encoding likewise provides a useful inductive bias without guaranteeing invariance to montage or reference changes. Recent audits document subject-related structure and limitations of robustness and interpretability in EEG foundation models [11, 14]. Physiological analyses also provide positive evidence, including sensitivity to electrophysiological components such as alpha activity [12]. The unresolved issue is therefore the strength and transferability of physiological evidence, rather than its complete absence from the literature.

1.3 Scope and Research Questions

Our data preparation spans 14 corpora covering several EEG paradigms and clinical applications. The experiments reported here have a narrower scope: we compare cross-subject BCI2A decoding and montage sensitivity, and inspect REVE geometry on BCI2A, FACED, and PhysioNet motor imagery. We ask whether class information is accessible in frozen features, how readout and montage choices affect decoding, and whether latent organization supports a physiological interpretation beyond subject-associated variation.

The contribution is an empirical audit with explicit limits on these inferences. We separate evaluation protocols from results, compare against EEGNet, and combine predictive scores with descriptive diagnostics.

We also developed a hierarchical encoder with a training objective based on LeJEPA (Section A). However, within the time available for this internship, we were unable to bring this training to convergence. No downstream results are therefore reported for this architecture. This limitation leaves its effectiveness unresolved and does not establish that LeJEPA is unsuitable for EEG.

2 Data and Evaluation Methods

2.1 Data Coverage and Experimental Scope

We processed 14 corpora spanning clinical and experimental EEG tasks (Table 1). TUEG supplies the self-supervised pretraining data prepared for the proposed encoder, with TUAB and TUAR subsets excluded. This exclusion describes our prepared pretraining corpus; it does not establish the training-data composition of the external pretrained models. Their corpus overlaps require a separate model-specific audit.

This breadth concerns tasks and recording contexts within EEG, rather than joint learning across independent sensor modalities such as EEG and MEG. Sleep-EDF originates from polysomnography, but the present analysis does not evaluate multimodal fusion. Likewise, typing-aligned METASPEECH epochs should not be interpreted as equivalent to covert-speech trials.

Quantitative model comparisons are restricted to the four-class BCI Competition IV dataset 2a, abbreviated BCI2A [3]. REVE cosine diagnostics additionally cover FACED [4] and PhysioNet MI [13]; t-SNE and PCA diagnostics use BCI2A. The broader corpus coverage establishes the scope of data preparation and future evaluation, not evidence of successful transfer across every listed task.

2.2 Models, Splits, and Reporting

The eight foundation models are evaluated with pretrained backbone weights. EEGNet is trained from scratch as a supervised baseline. The pooled SVM uses seven training and two test subjects; the token-preserving head and montage ablation use seven training, one validation, and one test subject. These distinct protocols answer related questions but do not form a controlled head-only ablation. Each decoding score

Table 1: Scope of data preparation. All entries concern EEG or the EEG component of a broader recording protocol. Inclusion denotes processing coverage, not a completed numerical comparison on every corpus.

Task	Corpora	Role or recording context
Self-supervision	TUEG	Large-scale clinical EEG; TUAB and TUAR subsets excluded from the prepared pretraining data.
Pathology detection	TUAB	Normal versus abnormal clinical recordings.
Sleep staging	Sleep-EDF	EEG from polysomnography with sleep-stage labels.
Seizure detection	CHB-MIT; Siena	Pediatric and adult scalp EEG, respectively.
Motor imagery	BCI IV-2a; PhysioNet MI; SHU-MI	Motor-imagery paradigms; SHU-MI provides 32-channel recordings.
Emotion recognition	SEED-V; FACED	Movie-elicited emotional responses and affective EEG.
Imagined speech	BCI 2020 (Track 3); KaraOne; CHISCO	Imagined speech; phonemes and words; Chinese reading and recall phases.
Speech-related typing	METASPEECH	EEG during typing, with epochs aligned to individual keystrokes.

describes one run, without repeated-split uncertainty estimates.

Balanced accuracy is the mean recall across classes and is the primary comparison metric; its four-class chance reference is 25%. Accuracy and macro-F1 provide complementary summaries. Numerical differences are reported descriptively, without significance claims. Diagnostic bootstrap intervals concern subjects, while PCA dispersions concern overlapping folds; neither supplies uncertainty for the separate decoding results.

2.3 Average-Pooled Linear SVM

For each frozen feature extractor, all intermediate token dimensions are averaged while preserving the final feature dimension. Thus, a tensor $H \in \mathbb{R}^{B\times C\times L\times D}$ is reduced to

$$\bar{h}_b = \frac{1}{CL}\sum_{c=1}^{C}\sum_{l=1}^{L} H_{bcl:} \in \mathbb{R}^D. \tag{1}$$

Here B is the batch size, C the channel count, L the number of temporal tokens, and D the feature width. Pooling removes explicit channel and temporal indexing. A linear support vector machine (SVM) is fitted after feature-wise standardization, with the mean and standard deviation estimated from training examples only. The regularization parameter is fixed at 1, with inverse-frequency class weights. No supervised adaptation of the feature extractor or input channel projections is performed in this protocol.

All runs use mapped channels and a subject-disjoint split with seven training subjects and two test subjects (random seed 100). No validation set or hyperparameter search is used for this comparison. The reported metrics summarize test epochs pooled across the two held-out subjects; they are not subject-level means or leave-one-subject-out estimates.

2.4 Common Token-Preserving Readout

Common readout. All pretrained models use the same readout design, adapted to each backbone's output shape, with separately learned head parameters. Pretrained backbone weights remain fixed throughout supervised training. Dropout and other training-dependent operations within the backbone remain in evaluation mode.

Let $H_b = (h_{b1}, \ldots, h_{bS})$ be the latent tokens of an example, with $h_{bs} \in \mathbb{R}^D$. Channel, patch, or summary-token axes are combined into the index s, without averaging their representations. A shared affine projection reduces each token to $q = 32$ features, followed by concatenation and an affine classifier with K outputs:

$$u_{bs} = P\ \mathrm{Dropout}_{0.2}(h_{bs}) + a, \tag{2}$$

$$\ell_b = W(u_{b1}\| \cdots \| u_{bS}) + b, \tag{3}$$

Here $\|$ denotes concatenation, $P \in \mathbb{R}^{32\times D}$, and $W \in \mathbb{R}^{K\times 32S}$. The row-wise weight norms of these two maps are constrained to at most 1 and 0.25, respectively.

There is no nonlinear activation between the two maps. With dropout disabled at evaluation, the logits are therefore affine in the flattened latent features. This is a factorized linear probe: all token positions share the same 32-dimensional feature projection. It is more constrained than an unrestricted linear classifier on all SD features. A common head design also does not imply equal capacity: its parameter count is $32(D+1) + K(32S+1)$ and varies with the backbone's token count and width.

Input adaptation. EEGPT and LaBraM additionally use a learned 1×1 channel convolution before the backbone. Freezing the backbone leaves this adapter trainable, and the optimizer updates it together with the head. Their protocol is thus frozen-backbone probing with input adaptation, rather than probing a completely fixed feature extractor. This distinction must accompany the results, since changes to the adapter

can change the representations even though pretrained weights remain fixed.

Training and evaluation. Subjects are assigned to disjoint training, validation, and test sets using target proportions of 80%, 10%, and 10%, with at least one subject in each held-out set. For BCI2A this gives seven training subjects, one validation subject, and one test subject. A random seed of 100 is used for both the subject split and training randomization. This is a single subject-disjoint split, not leave-one-subject-out cross-validation.

For the pretrained models, training uses cross-entropy without label smoothing, AdamW with learning rate 4×10^{-4} and weight decay 0.01, batches of 64, and gradient-norm clipping at 1. Training lasts at most 50 epochs, with early stopping after ten epochs without improvement in validation balanced accuracy. The best validation checkpoint is restored before test evaluation. Accuracy, balanced accuracy (mean class recall), and macro-F1 are recorded; balanced accuracy is the selection criterion.

EEGNet is a separate supervised baseline trained without pretrained weights. Its backbone is optimized together with its classifier, so its results must be distinguished from those obtained with frozen pretrained representations. Its training uses AdamW with learning rate 10^{-3}, weight decay 10^{-4}, and dropout 0.25, with the same subject split and validation-based selection criterion.

2.5 Montage Transformation

The montage comparison assesses the effect of replacing the native channel layout with a fixed target layout. Here, *raw* denotes native channels from the preprocessed dataset, not unprocessed EEG. The mapped condition uses the nearest-source rule illustrated in Figure 1. For target electrode i, the closest source electrode $j^*(i)$ supplies the complete channel signal if its distance is at most r; otherwise the target channel is filled with zeros.

This operation copies signals rather than interpolating several electrodes. Different target locations can select the same source, and sources that are never selected are omitted. Mapping therefore changes spatial sampling and may introduce duplicate or missing channels. Its effect is not limited to channel ordering, and improved performance would not by itself demonstrate montage invariance or recovery of unobserved activity.

We evaluate the frozen CBraMod and EEGMamba backbones with the common head, alongside EEGNet trained end to end. Each condition uses seven training subjects, one validation subject, and one test subject. The intended control keeps the subject split, underlying windows, labels, checkpoint, and training settings fixed within each model. Mapping can nevertheless change the readout size through the channel count, so this comparison measures the effect of the complete montage-processing pipeline.

2.6 Latent-Space Diagnostics

All descriptive figures use REVE with mapped channels. They are not cross-model evidence. Pooling and projection choices differ by analysis and are specified below so that their conclusions are not conflated.

t-SNE. The t-SNE analysis [15] uses 1,800 BCI2A epochs from REVE, sampled without replacement at 200 epochs per subject across all nine subjects. Sampling is balanced by subject, not explicitly by class. Token features are averaged over the channel and temporal axes, yielding one 512-dimensional vector per epoch. Features are standardized across the visualization sample, reduced to 50 principal components, and embedded with perplexity 30, PCA initialization, and 1,000 optimization iterations (random seed 100).

Within-subject cosine similarity. To examine class-related angular structure while conditioning on subject, we compare distinct pairs of epochs from the same participant. REVE tokens are averaged over all axes except batch and feature dimension, then ℓ_2-normalized without centering. For pooled features h_a, h_b, the similarity is $s_{ab} = h_a^\top h_b / (\|h_a\|_2 \|h_b\|_2)$. The retained samples contain 2,997 epochs from nine BCI2A subjects, 2,952 from 123 FACED subjects, and 2,943 from 109 PHYSIONETMI subjects.

For subject u, define

$$\Delta_u = \overline{s}_{u,\text{same}} - \overline{s}_{u,\text{different}}, \qquad \overline{\Delta} = \frac{1}{U} \sum_{u=1}^{U} \Delta_u. \quad (4)$$

U is the number of retained subjects. Each subject's means use all eligible pairs among their retained epochs. The reported 95% percentile bootstrap interval resamples subjects 5,000 times, rather than treating the many dependent pairs as independent observations. Histograms use equal numbers of same-class and different-class pairs within each subject, with a common per-subject cap derived from a 50,000-pair budget per condition. Subjects with fewer eligible pairs contribute fewer histogram samples.

Training-only PCA. PCA retains all REVE tokens as a single flattened vector, whereas t-SNE and cosine similarity use pooled features. The BCI2A analysis uses nine outer folds, each holding out one test subject and the next subject in cyclic order for validation. Ten principal components are fitted to centered, unstandardized features from the remaining seven subjects. The 512-epoch training budget allocates 73 epochs per subject, giving 511 sampled training epochs per fold. Neither validation nor test examples enter the PCA fit.

The displayed fold reserves A01 for testing and shows validation subject A02. RGB colors encode the first three component scores after scaling each component by its training-set 1st and 99th percentiles and clipping to $[0, 1]$. Colors are therefore an unsupervised summary of three latent coordinates, not predicted

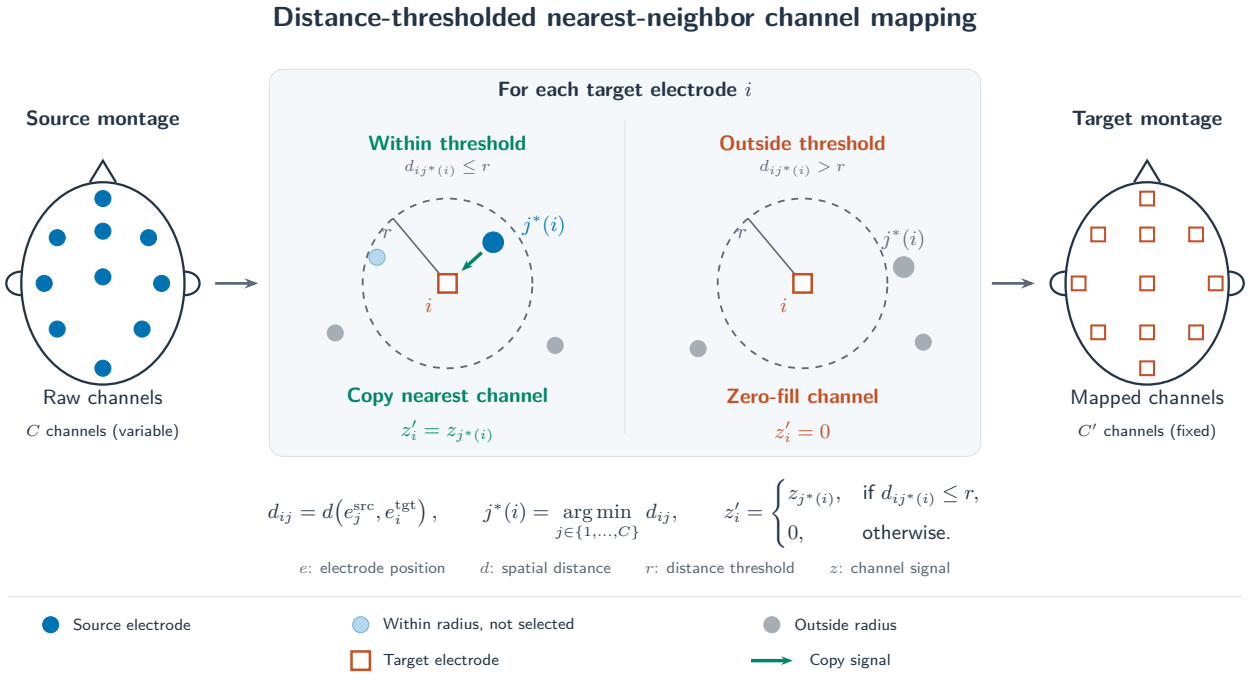


Figure 1: Channel-mapping rule for the montage ablation. Each fixed target electrode independently selects the closest source within a distance threshold; targets without an eligible source are zero-filled. The symbols z and z' denote channel signals in this schematic, not the latent tokens used elsewhere in the report. This figure describes the transformation and contains no downstream performance comparison.

labels, physiological categories, or a measure of confidence. The color transform is also fitted without validation or test data.

3 Results

The following comparisons use the protocols in Section 2. We first report decoding and montage results, then examine the geometry of REVE representations. Tables retain the individual measured scores; percentage-point differences in the text are rounded to two decimals.

3.1 Linear SVM Probing of Average-Pooled Features

Table 2 reports balanced accuracy ranging from 26.80% to 33.31%. EEGPT obtains the lowest score and REVE the highest. CBraMod achieves 30.08%, while the other models lie between 27.72% and 28.72%. These estimates are numerically above the 25% four-class balanced-accuracy chance reference, but the absence of repeated splits or a permutation test prevents a significance claim. Macro-F1 follows the same broad pattern, ranging from 0.263 to 0.325.

After average pooling, cross-subject linear discriminability is weak. This does not demonstrate complete class inseparability: the scores exceed the chance reference numerically, and averaging can remove informative spatial or temporal contrasts. The supported conclusion is that this particular summary exposes relatively little class information to a linear SVM on the held-out subjects, not that the complete latent representations contain no task information.

Table 2: Linear SVM on average-pooled, frozen features for mapped BCI2A. All metrics are on a 0–1 scale. The split contains seven training and two test subjects.

Model	Accuracy	Bal. acc.	Macro-F1
CBraMod	0.300636	0.300846	0.299699
EEGMamba	0.278837	0.278865	0.277760
CSBrain	0.279746	0.279834	0.279020
LuMamba	0.287012	0.287205	0.287390
LUNA	0.286104	0.286117	0.286464
LaBraM	0.277021	0.277174	0.273225
EEGPT	0.267938	0.268049	0.262756
REVE	0.333333	0.333063	0.325038

3.2 Frozen-Backbone Evaluation with a Common Head

REVE leads the pretrained models at 65.21% balanced accuracy. The next best scores are 53.47% for EEGPT and 42.60% for LaBraM. Other pretrained models score between 31.01% and 37.29% (Table 3). EEGNet reaches 76.77%, exceeding REVE by 11.56 percentage points, and also has the highest accuracy and macro-F1 and the lowest cross-entropy in this experiment. This is evidence for a strong task-specific supervised baseline under the evaluated conditions, not a comparison with

Table 3: Mapped BCI2A evaluation with seven training, one validation, and one test subject. Pretrained models use the common token-preserving head with frozen backbone weights; EEGPT and LaBraM also learn input channel adapters. EEGNet is trained end to end with its own classifier. Epoch denotes the checkpoint selected by validation balanced accuracy. Accuracy, balanced accuracy, and macro-F1 are on a 0–1 scale; loss is test cross-entropy.

Model	Epoch	Accuracy	Bal. acc.	Macro-F1	Loss
CBraMod	21	0.366483	0.366907	0.352562	1.313258
EEGMamba	3	0.348066	0.350022	0.323536	1.326661
CSBrain	6	0.372007	0.372859	0.368457	1.301348
LuMamba	4	0.309392	0.310068	0.297328	1.365851
LUNA	14	0.346225	0.347750	0.320760	1.361484
LaBraM	19	0.427256	0.425955	0.424304	1.213312
EEGPT	26	0.534070	0.534666	0.535005	1.051783
REVE	11	0.651934	0.652083	0.654002	0.919220
EEGNet	49	0.767956	0.767696	0.767876	0.618744

fully fine-tuned foundation models.

For every pretrained model, the common-head score is numerically higher than its pooled-SVM score. This cannot be attributed solely to the readout: the test subjects and selection procedures differ, and EEGPT and LaBraM additionally adapt their inputs. In particular, the two affine projections do not make the common head a nonlinear classifier of fixed features. A matched split with identical adaptation rules is required to isolate the effect of retaining individual tokens instead of averaging them.

3.3 Channel-Mapping Ablation: Raw versus Mapped Montages

Native channels improve both pretrained models (Table 4). CBraMod increases from 36.69% to 40.56% balanced accuracy, a gain of 3.87 percentage points. EEGMamba increases from 35.00% to 40.00%, a gain of 5.00 points. Accuracy and macro-F1 also improve, while cross-entropy decreases. The consistency across these metrics supports a raw-montage advantage for these two runs, without establishing its reproducibility across participants.

EEGNet shows the opposite pattern: mapped channels yield 75.30% balanced accuracy versus 63.91% with raw channels, a gain of 11.39 points. Accuracy and macro-F1 likewise favor mapping, and loss falls from 0.925231 to 0.650881. EEGNet outperforms both pretrained models within each montage condition. Its mapped score differs from the 76.77% in the common-head comparison; the two runs are retained separately and are not averaged or treated as controlled replicates.

Interpretation. These opposite effects argue against a universal benefit of channel mapping. Retaining native observations may benefit the pretrained models, whereas a fixed layout may help the supervised baseline; these mechanisms remain hypotheses. Mapping also omits, duplicates, or zero-fills channels, and the selected epochs differ between conditions. The results therefore do not isolate geometry from information loss, model capacity, or optimization effects. The distance threshold, target montage, coverage statistics, and repeated subject-level comparisons are still needed for a more complete account of these differences.

3.4 Subject and Class Organization in t-SNE

In Figure 2, the left panel contains recognizable subject-enriched regions, particularly for A02, A05, A07, and A09, alongside substantial overlap elsewhere. Coloring the same coordinates by task class reveals no comparably distinct class regions. Subject membership is thus more visually apparent than class membership in this particular projection. This observation motivates controlling subject identity in subsequent analyses and keeping subjects separate during downstream evaluation.

The plot does not quantify how much subject or class information is present in the full representation. Averaging tokens can discard task information, and a nonlinear two-dimensional embedding can obscure class separation available in other directions. Nor can subject-associated structure distinguish physiology from session or acquisition effects. Consequently, these clusters provide descriptive evidence of subject-related organization, not proof of artifact dependence or failed cross-subject decoding.

3.5 Within-Subject Cosine Similarity

The distributions overlap strongly (Figure 3). For BCI2A, the mean subject gap is about 0.0001; its displayed interval is $[0.0000, 0.0001]$. The lower endpoint is rounded, so its printed value is insufficient to decide whether the unrounded interval excludes zero. FACED and PHYSIONETMI have larger positive gaps, 0.0021 $[0.0011, 0.0032]$ and 0.0010 $[0.0006, 0.0015]$, respectively. These positive intervals support a small average class association in angular similarity within the sampled subjects, not a decoding accuracy.

Absolute similarities are high for both pair types, with displayed means between approximately 0.957

Table 4: Mapped-versus-raw BCI2A ablation with a 7/1/1 subject split. CBraMod and EEGMamba use the common head with a frozen backbone; EEGNet uses its own classifier and is trained end to end. Epoch is selected on validation balanced accuracy. Metrics use a 0–1 scale, except cross-entropy loss. The EEGNet mapped run is distinct from the run in Table 3.

Model	Montage	Epoch	Accuracy	Bal. acc.	Macro-F1	Loss
CBraMod	Mapped	21	0.366483	0.366907	0.352562	1.313258
CBraMod	Raw	8	0.405157	0.405605	0.402207	1.300800
EEGMamba	Mapped	3	0.348066	0.350022	0.323536	1.326661
EEGMamba	Raw	10	0.397790	0.400017	0.371253	1.279251
EEGNet	Mapped	32	0.753223	0.753001	0.753213	0.650881
EEGNet	Raw	11	0.637201	0.639129	0.628166	0.925231

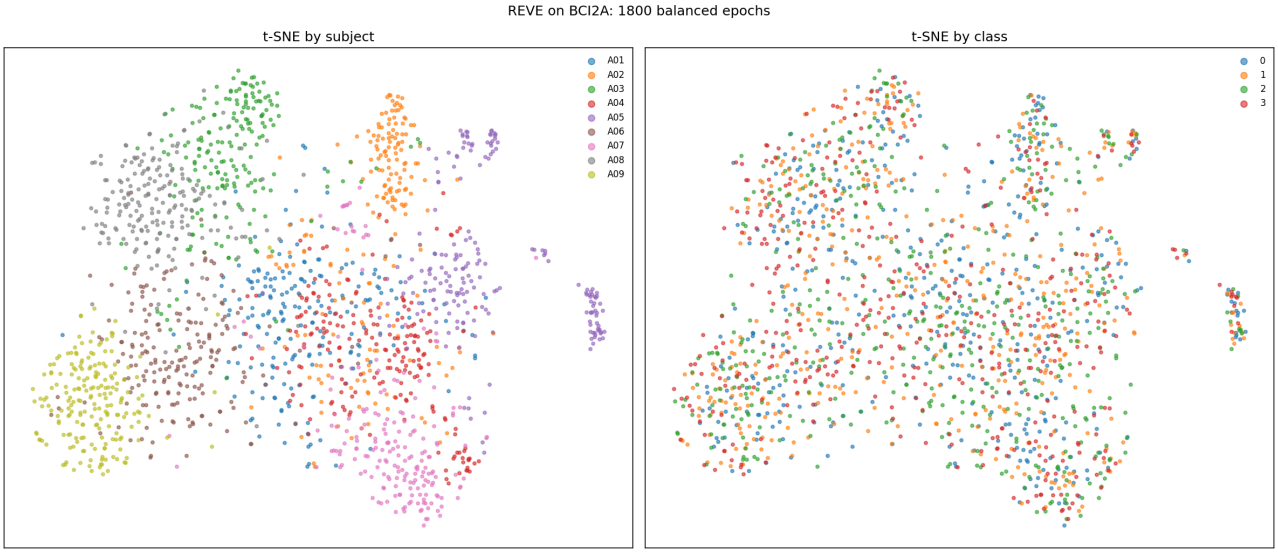


Figure 2: REVE representations of 1,800 mapped BCI2A epochs in one t-SNE embedding. Identical coordinates are colored by subject (left) and class (right). Several neighborhoods are enriched for individual subjects, while class labels are broadly mixed. Standardization, PCA, and t-SNE are fitted on the displayed sample for descriptive analysis; this is not a held-out classification result.

and 0.987. This indicates strongly aligned pooled vectors relative to the origin. A shared mean component can produce such alignment because features were not centered. High cosine similarity therefore does not establish representation collapse, and a small mean gap does not exclude informative low-variance directions. The subject-wise summary also differs from the histogram mean difference: for example, FACED's pair-weighted means differ by 0.0036, while its equally weighted subject gap is 0.0021.

3.6 PCA across Subjects and Epoch-Level Signal Inspection

Variance and stability. The cumulative top-ten training explained variance is $21.20\% \pm 0.31\%$ across folds. Subspace stability is assessed by refitting PCA on five random 80% subsets of each fold's training sample. For reference and refitted orthonormal loading matrices V and $\widetilde{V}$, the score is $10^{-1}\sum_{k=1}^{10}\sigma_k^2(V\widetilde{V}^\top)$, the mean squared cosine of their principal angles. The mean fold score is 0.909 ± 0.014. These dispersions are fold standard deviations, not confidence intervals; folds share much of their training data.

The same components capture $14.08\% \pm 2.37\%$ of held-out test energy, measured relative to the *training* mean. This is not variance about the test subject's own mean and can include a between-subject mean shift. The lower capture than on training data is compatible with a shift in feature structure, but also with fitting PCA to maximize training variance. The results indicate a reasonably stable leading subspace under training subsampling while leaving most feature variation outside its ten dimensions. They do not measure class discrimination.

Class overlap and signal context. In the displayed fold, PC1 and PC2 account for 5.1% and 3.1% of training variance. The four validation classes overlap extensively in this plane (Figure 4a), and their average RGB colors are similar (Figure 4b). Together, these observations show little visual class separation in the leading coordinates for A02. Since the first two components explain only about 8.2% of training variance and PCA ignores labels, this does not rule

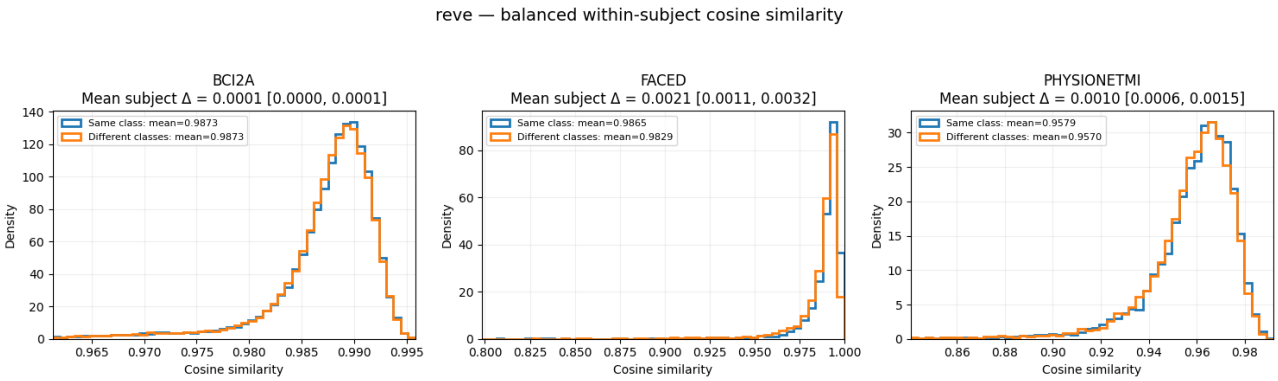


Figure 3: Within-subject cosine similarities of mean-pooled REVE representations with mapped channels. Blue and orange histograms compare same-class and different-class pairs. Titles report the equally weighted mean subject gap and its subject-bootstrap 95% interval; legend means refer to the sampled pairs and need not have the same weighting. The horizontal ranges differ across datasets and are restricted around the central 99% of sampled similarities.

out separation in later components or in the full token representation.

Figure 4c places the latent colors alongside eight reproducibly sampled epochs. Background colors vary between examples, including examples of the same class; the seventh displayed epoch contains particularly large transients on the plotted channel. This juxtaposition motivates testing associations with signal amplitude or artifacts, but it does not establish such an association. The embedding uses all channels, the trace shows only one, and neither artifact labels nor quantitative signal–latent correlations are included. Calling these transients ocular or muscular artifacts would therefore exceed the available evidence.

4 Discussion

Predictive accuracy and physiological evidence. These experiments distinguish successful prediction from evidence that a representation captures task-relevant physiological mechanisms. Neither classification accuracy nor a visually structured latent space identifies the source of the predictive signal. The present results do not establish that the pretrained features encode transferable physiological markers, independently of participant characteristics, acquisition conditions, or artifacts. This is a limitation of the evidence, rather than proof that such markers are absent from the representations.

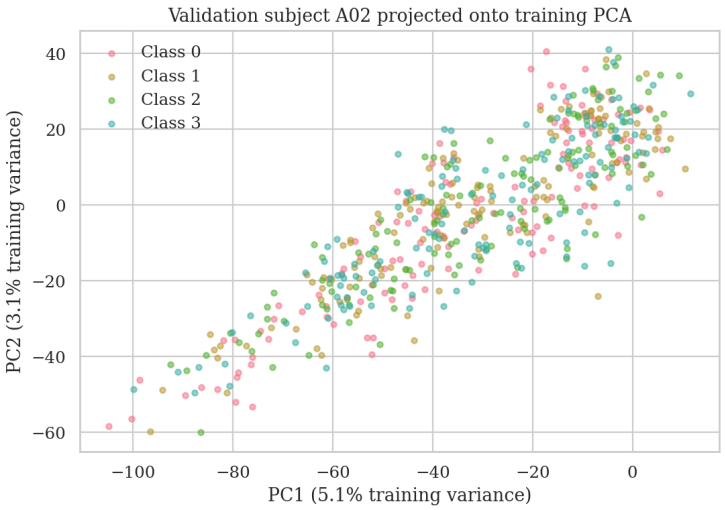
Physiological evidence in the broader literature is heterogeneous. For example, relevance analyses identify task-appropriate brain regions, although stable attribution maps can coexist with degraded predictions under perturbation [14]. Other work separates subject-related information from established physiological markers through targeted controls [11]. A separate study demonstrates representation sensitivity to alpha activity and individual electrophysiological variation [12]. Such findings preclude a blanket claim that foundation models have learned no physiological features. They do not, however, establish that the same mechanism supports reliable decoding across subjects and recording conditions.

Subject structure and transfer. The REVE t-SNE display contains more apparent subject organization than class organization. Together with the weak pooled-SVM results, this raises a concrete concern: learning may retain subject-dependent cues that do not transfer reliably to new participants. These cues could include artifacts, but also genuine individual physiology or session effects. The plot alone cannot identify their origin, quantify their dominance in the full space, or extend this finding to every evaluated model. Independent diagnostic work nevertheless supports treating subject identity as a serious potential shortcut in foundation-model evaluation [11].

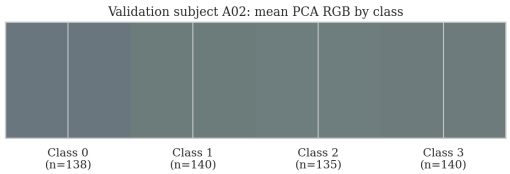
Subject-disjoint evaluation is necessary for testing transfer, but one held-out participant cannot characterize a population. Repeated splits or leave-one-subject-out evaluation, followed by cross-session and cross-dataset testing, would strengthen the conclusions. Subject-identity decoding and comparisons with artifact annotations would help determine whether an improvement reflects task information or a persistent nuisance variable. The observed montage sensitivity further shows that transfer depends on how recordings are presented to the model.

A compact supervised baseline. The evaluated EEGNet baseline has approximately 3,000 parameters, yet it outperforms every frozen foundation model in the mapped common-head comparison. Its balanced accuracy of 76.77% exceeds the best pretrained result, REVE at 65.21%, by 11.56 points. EEGNet also leads the montage ablation in both mapped and raw conditions. Thus, it is consistently the stronger baseline within the reported comparable experiments, despite its small size. This supports retaining compact, task-specific models as serious comparators [10].

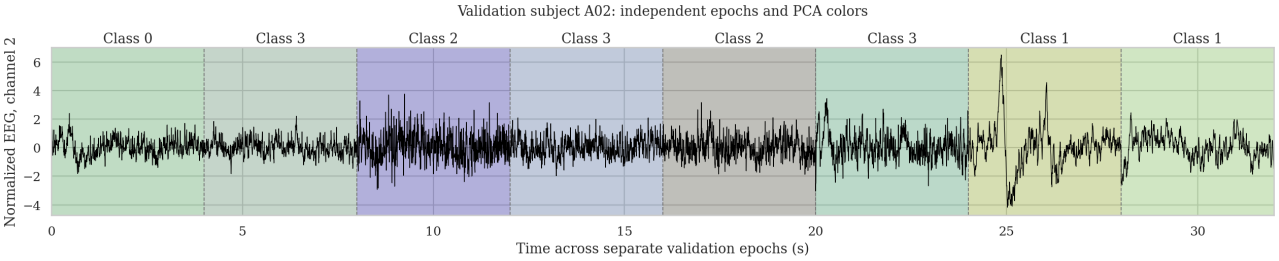
This advantage is conditional on the evaluated training regimes: EEGNet learns all its features from



(a) Validation epochs in the training PCA basis.



(b) Class means of the three-component RGB encoding.



(c) Eight separately sampled four-second epochs, with latent RGB backgrounds.

Figure 4: Training-fitted PCA of mapped REVE features, evaluated visually on BCI2A validation subject A02. The scatter and class-color means use all 553 validation epochs (138, 140, 135, and 140 in classes 0–3). In the signal panel, the trace is normalized input channel index 2; dashed boundaries separate independent epochs. The horizontal axis is cumulative display time, not one continuous EEG recording.

task labels, whereas the foundation backbones remain frozen. Their optimization settings also differ. The findings therefore do not establish that EEGNet always exceeds foundation models across datasets, tasks, or full fine-tuning protocols. They show that model scale and pretraining alone are insufficient to guarantee better decoding in this BCI2A evaluation.

Pooling, dimensionality, and interpretability. Average-pooled features yield only 26.80–33.31% balanced accuracy with a linear SVM, compared with a 25% chance reference. They consequently show weak transferable linear separation under this protocol. Pooling can remove contrasts between channels or time points, however, so these results cannot establish inseparability of all latent features. Prior work similarly identifies pooling as a limitation of frozen-feature assessment [14]. The higher token-preserving scores motivate a matched pooling ablation rather than a claim that nonlinear classification alone resolves the problem.

High-dimensional representations also complicate interpretation. Retaining more tokens increases the downstream parameter count and can improve the fit to incidental correlations when subjects are few. Some representation similarity measures have specific high-dimensional failure modes: for example, CCA-type statistics become uninformative when feature dimension exceeds the number of observations [9]. This does not imply that high dimensionality automatically inflates properly held-out accuracy, nor that the reported scores are artifacts. It motivates matched probe capacities, training-only dimensionality reduction, and shuffled-label or random-feature controls that respect the subject structure.

Accuracy should therefore be accompanied by evidence about what the model uses: perturbations of selected frequency bands and regions, sensitivity to physiological markers, robustness to artifact removal, and replication across subjects. The strongly aligned uncentered cosine vectors and the small leading-component class contrast are useful diagnostics, but neither is a direct measurement of physiological meaning. A favorable metric must be interpreted together with its feature transformation, effective capacity, and sampling unit.

Learning invariances with LeJEPA. LeJEPA offers a way to align representations of related views while regularizing their distribution. More precisely, it is a non-contrastive joint-embedding method: its anti-collapse mechanism uses SIGReg rather than explicit

negative pairs [1]. Contrastive self-supervision has separately shown strong results on time-series benchmarks, including multivariate data, as illustrated by TS2Vec [19]. LuMamba further evaluates LeJEPA on EEG and reports stronger transfer with a combined reconstruction and alignment objective than with either alone [2]. These results motivate hybrid as well as pure latent objectives; they do not establish physiological specificity or invariance to artifacts.

For an EEG application, a central research objective is to construct views that preserve the neural phenomenon of interest while varying nuisance components. Candidate augmentations include independently varying physiologically plausible ocular or muscle contamination, bounded sensor gain changes, and spatially constrained channel masking. Where paired contaminated and artifact-reduced signals are available, they could provide an additional invariance test. These are proposed experiments, not outcomes established by the present benchmark.

The transformations must be validated for the task: aggressive filtering, amplitude changes, or temporal warping may remove neural information along with artifacts. Augmentations should vary nuisance content independently between views; an artifact preserved in both views can itself satisfy the alignment objective. Equally, suppressing all subject information is not an appropriate universal target, since some individual physiological variation may be relevant. Useful invariance must be demonstrated through artifact robustness together with preserved marker sensitivity and held-out task performance.

Limitations and priorities for validation. The number of prepared corpora exceeds the coverage of the reported experiments. Eight-model decoding results on one task and diagnostic figures from one encoder cannot establish universal properties of EEG foundation learning. External models also differ in pretraining data, architecture, representation size, and input adaptation, so this audit cannot causally rank their pretraining objectives. Single-run scores and one held-out test subject in the common-head comparison limit population-level claims. The two EEGNet mapped runs are distinct observations, not a replication estimate.

A stronger evaluation should cross readout type and pooling under identical subject splits, control trainable adapters and classifier capacity, and compare frozen, fine-tuned, and randomly initialized encoders. Multiple seeds and held-out subjects should accompany subject-level uncertainty estimates. Artifact annotations, physiological marker tests, and cross-session or cross-corpus transfer would then address whether gains reflect useful invariances. Incomplete reporting of dataset versions, exact split identities, preprocessing choices, montage specifications, and potential pretraining overlap further limits reproducibility and attribution of performance differences.

5 Conclusion

The central challenge for EEG foundation models is to learn latent representations that preserve relevant multivariate dynamics while remaining useful across subjects and recording conditions. In the present BCI2A experiments, compact supervised EEGNet outperforms all frozen foundation backbones under the compared protocols. Average-pooled SVM features show weak transferable class discrimination, and montage changes have model-dependent effects. REVE diagnostics reveal subject-associated organization without identifying its physiological or artifactual origin. These observations qualify the evidence for general-purpose representations; they do not establish their impossibility.

No learning objective can be selected as universally best from these data. Signal reconstruction retains detailed observations but need not prioritize physiological mechanisms. Latent alignment can encourage invariance while also preserving nuisance factors shared between views. Joint spatial and temporal modeling, meaningful prediction targets, and carefully validated augmentations should therefore be studied together. LeJEPA and hybrid reconstruction–alignment objectives are relevant candidates, provided that the intended invariances preserve task-related neural information. The additional encoder specification in Section A is a proposal for such experiments, not an empirically supported solution.

The breadth of the prepared EEG corpora makes evaluation across clinical, affective, motor, sleep, and speech-related settings possible, but that broader validation remains to be reported. Progress should be judged by reproducible cross-subject and cross-dataset transfer, competitive compact baselines, and direct evidence about the physiological content of latent features. Model size, low reconstruction error, or favorable geometric metrics alone are insufficient criteria for useful foundation learning.

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A Proposed Encoder and Joint-Embedding Training

This appendix specifies an additional hierarchical EEG encoder and a candidate training procedure. None of the quantitative results in Section 3 evaluates this architecture. The design is retained to make the proposed research direction precise; its components, augmentations, and objective require controlled ablation before any performance or physiological claim can be made. It is distinct from the eight pretrained models compared in the main text.

The encoder combines electrode-wise tokenization, spatial mixing, and bidirectional temporal processing. It returns channel-resolved tokens and a global representation. The proposed global objective follows LeJEPA-style view agreement and distributional regularization without waveform reconstruction. The specification below defines a patch-based configuration, rather than a validated optimal architecture.

Let $X \in \mathbb{R}^{B\times C\times T}$ denote a batch of EEG windows, where B is the batch size, C the number of channels in a fixed canonical order, and T the number of time samples. Electrode coordinates are denoted by $P = (p_1, \dots, p_C)^\top \in \mathbb{R}^{C\times 3}$. A presence code $a_{bc} \in \{0, 1, 2\}$ accompanies each channel; code zero marks an absent channel, and both nonzero codes are treated as available by the attention masks. We write $m_{bc} = \mathbb{1}[a_{bc} \neq 0]$ and assume that each window contains at least one available channel. Hereafter, $\|$ denotes concatenation, $\odot$ element-wise multiplication, and RMSNorm and LayerNorm denote root-mean-square normalization and layer normalization, respectively. Features are column vectors, and linear maps act on their last dimension. The presence input is labeled p in the figures and a in the text.

A.1 Model Pipeline Overview

Figure 5 summarizes the mapping

$$(Z, g) = f_\theta(X, a; P), \tag{A.1}$$

where $Z \in \mathbb{R}^{B\times C\times N_4\times D_z}$ contains the token embeddings and $g \in \mathbb{R}^{B\times D_z}$ contains the global embeddings. These correspond to the outputs z and z_{glob} in the figures. The channel axis is preserved throughout the encoder; only temporal resolution and feature dimension change. The input length T is variable; the encoder uses $D_z = 384$. At a sampling rate of 200 Hz, an input window spans $T/200$ seconds.

Signal tokenization. The stem first applies a signed logarithmic compression,

$$\widetilde{X} = \operatorname{sign}(X) \odot \log(1 + |X|). \tag{A.2}$$

Each channel is processed independently by the same two temporal convolutional layers. Both have kernel size nine and are followed by eight-group GroupNorm and SiLU. The first convolution maps one input feature to $d_s = 32$ features; the second preserves this width and uses eight convolution groups. Padding preserves the time dimension. The resulting features are partitioned into non-overlapping patches of $p = 20$ samples. With $N_1 = \lfloor T/p \rfloor$, the flattened patch vector $r_{bcn} \in \mathbb{R}^{pd_s}$ is embedded as

$$H^{(0)}_{bcn} = W_{\text{stem}} \operatorname{LayerNorm}(r_{bcn}) + e_c + e_{a_{bc}}, \tag{A.3}$$

where $W_{\text{stem}} \in \mathbb{R}^{128\times 640}$, e_c is a learned channel embedding, and $e_{a_{bc}}$ is a learned presence embedding. Any incomplete final patch is discarded.

Hierarchical encoding. The four stages use feature widths $(D_1, D_2, D_3, D_4) = (128, 192, 256, 384)$ and block counts $(L_1, L_2, L_3, L_4) = (2, 2, 3, 3)$. Between stages, adjacent tokens are concatenated, normalized, and projected:

$$r^{(i)}_{bcn} = H^{(i)}_{bc,2n-1} \| H^{(i)}_{bc,2n}, \tag{A.4}$$

$$H^{(i+1,\text{in})}_{bcn} = W^{(i)}_{\downarrow} \operatorname{RMSNorm}(r^{(i)}_{bcn}), \tag{A.5}$$

where $W^{(i)}_{\downarrow} \in \mathbb{R}^{D_{i+1}\times 2D_i}$ and $n = 1, \dots, \lfloor N_i/2 \rfloor$. An unpaired final token is discarded. Thus, for patch size $p = 20$, the temporal lengths satisfy

$$N_i = \left\lfloor \frac{T}{p\, 2^{i-1}} \right\rfloor, \qquad i = 1, \dots, 4. \tag{A.6}$$

In particular, $N_4 = \lfloor T/160 \rfloor$; we require $T \geq 160$ so that at least one temporal token reaches the bottleneck.

Token bottleneck and global readout. The final hidden tensor is normalized and optionally projected:

$$Z_{bcn} = W_z \operatorname{RMSNorm}(H^{(4)}_{bcn}). \tag{A.7}$$

Here W_z is the identity because $D_z = D_4$; a learned projection is used when these dimensions differ. To obtain g_b, the CN_4 tokens are flattened into a sequence $(z_{bj})_j$. For each of $A = 8$ attention heads, linear maps produce keys and values of width $d_z = D_z/A = 48$:

$$k_{bhj} = \operatorname{RMSNorm}(W^K_h z_{bj}), \qquad v_{bhj} = W^V_h z_{bj}. \tag{A.8}$$

Each head has $Q = 4$ learned queries $q_{hq} \in \mathbb{R}^{d_z}$. A token inherits the availability of its channel, denoted by m_{bj}. The pooled head outputs are

$$\alpha_{bhqj} = \frac{m_{bj} \exp(q^\top_{hq} k_{bhj}/\sqrt{d_z})}{\sum_r m_{br} \exp(q^\top_{hq} k_{bhr}/\sqrt{d_z})}, \tag{A.9}$$

$$o_{bhq} = \sum_j \alpha_{bhqj} v_{bhj}. \tag{A.10}$$

Concatenating the heads for each query and then the queries gives $o_b \in \mathbb{R}^{QD_z}$. The global representation is

$$g_b = W_{\text{out}} W_{\text{merge}} \operatorname{RMSNorm}(o_b), \tag{A.11}$$

with $W_{\text{merge}} \in \mathbb{R}^{D_z\times QD_z}$ and $W_{\text{out}} \in \mathbb{R}^{D_z\times D_z}$. Thus, the global readout aggregates both channel and temporal information while excluding unavailable channels as attention keys. Training uses g_b directly; the token output Z remains available for downstream analysis.

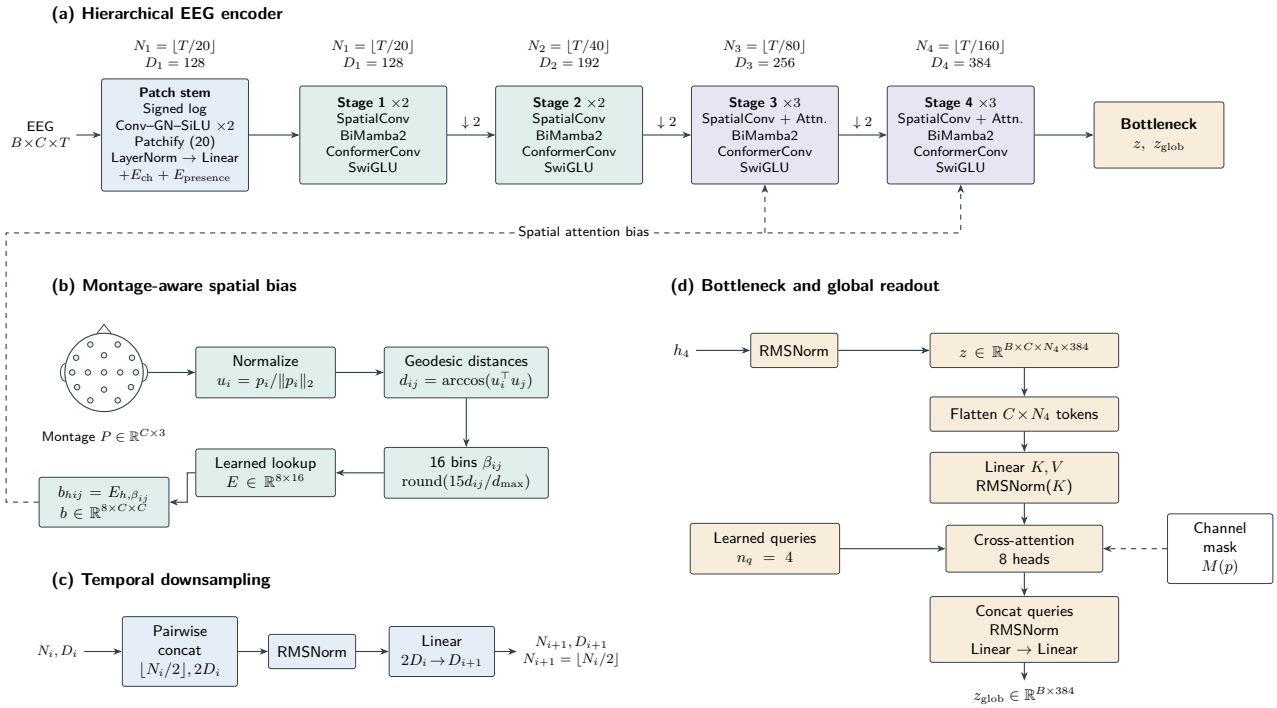


Figure 5: Model architecture. A patch stem feeds four encoder stages with progressively shorter temporal sequences and larger feature dimensions. Geodesic distances between electrode coordinates are quantized into 16 bins and mapped to learned attention biases in the last two stages. The bottleneck returns token embeddings z and a global embedding z_{glob} obtained with four learned queries. The input has shape $B \times C \times T$; stage lengths are $N_i = \lfloor T/(20 \cdot 2^{i-1}) \rfloor$ for $i = 1, \ldots, 4$. Feature widths correspond to the proposed architecture.

A.2 Encoder Architecture

Every block preserves its input shape $B \times C \times N \times D$. As shown in Figure 6, its four residual updates take the form

$$U_r = U_{r-1} + \text{DP}_\rho\big(\gamma_r \odot F_r(\text{RMSNorm}_r(U_{r-1}))\big), \tag{A.12}$$

for $r = 1, \ldots, 4$, with F_r denoting spatial mixing, temporal Mamba2, Conformer convolution, and SwiGLU, respectively. Each branch has its own normalization parameters and feature-wise LayerScale vector $\gamma_r \in \mathbb{R}^D$, initialized to 10^{-4}. During training, DropPath multiplies a branch by $\xi/(1-\rho)$, with $\xi \sim \text{Bernoulli}(1-\rho)$ independently sampled per batch example and branch. The rate ρ increases linearly from zero to 0.15 over the ten blocks. Ordinary dropout is zero in the reference architecture.

Convolutional spatial mixing. Let U denote the input to the spatial module, already normalized by the outer residual branch. SpatialConv applies an additional RMSNorm and masks unavailable input channels:

$$G_{bcnd} = m_{bc}\,[\text{RMSNorm}(U_{bcn})]_d. \tag{A.13}$$

At every temporal and feature index, two 1×1 convolutions mix the channel axis:

$$S^{\text{conv}}_{b,:,n,d} = W_{c,2}\,\text{SiLU}(W_{c,1} G_{b,:,n,d}), \tag{A.14}$$

where $W_{c,1}, W_{c,2} \in \mathbb{R}^{C\times C}$. This branch is present in all stages and uses a fixed channel ordering.

Montage-dependent attention bias. Coordinates p_i are taken from the standard 10–20 montage for the mapped channel names. We normalize them and compute angular distances:

$$u_i = \frac{p_i}{\max(\|p_i\|_2, \varepsilon)}, \tag{A.15}$$

$$d_{ij} = \arccos\big(\text{clip}(u_i^\top u_j, -1, 1)\big)\,. \tag{A.16}$$

The distances are discretized into $K_d = 16$ bins,

$$\beta_{ij} = \text{round}\left(\frac{(K_d - 1) d_{ij}}{\max(\max_{r,s} d_{rs}, \varepsilon)}\right), \tag{A.17}$$

where the denominator is clamped from below by $\varepsilon = 10^{-8}$. The resulting integer indices remain fixed during training. A learned table $E \in \mathbb{R}^{A\times K_d}$, initialized to zero, defines the head-specific bias $b_{hij} = E_{h,\beta_{ij}}$. This table is shared across the attention blocks of the encoder.

Statistics-conditioned spatial attention. Stages three and four add a beamformer-style attention branch to Equation (A.14). For each channel, queries and keys

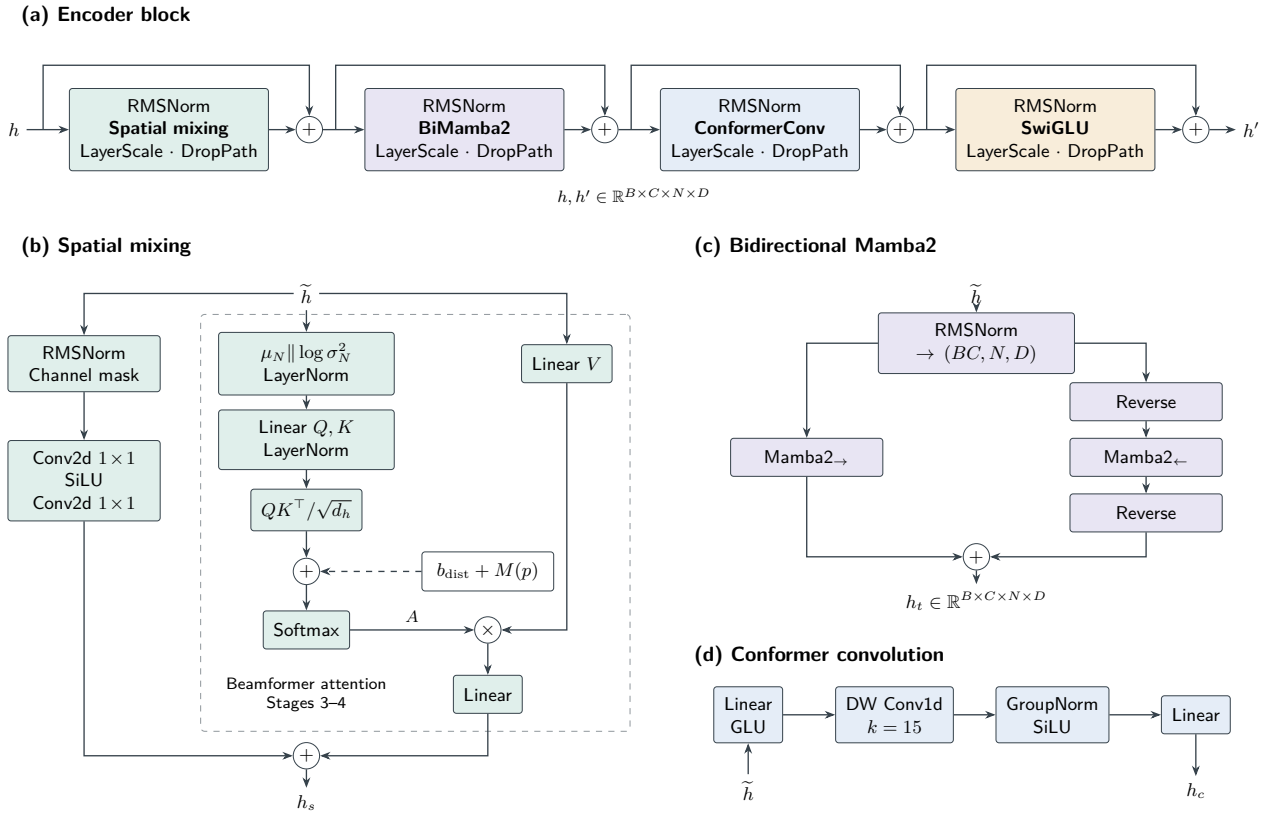


Figure 6: Encoder block and its constituent operations. Four pre-normalized residual branches perform spatial mixing, bidirectional Mamba2 processing, local temporal convolution, and a gated feed-forward transformation. Spatial attention is added to the convolutional spatial branch in stages three and four. $\widetilde{h}$ denotes a branch input after its outer normalization; $M(p)$ in the diagram is the additive channel-availability mask associated with the presence codes a in the text.

are conditioned on statistics over the temporal tokens:

$$\mu_{bc} = \frac{1}{N}\sum_{n=1}^{N} U_{bcn}, \tag{A.18}$$

$$v_{bc} = \frac{1}{N}\sum_{n=1}^{N} (U_{bcn} - \mu_{bc})^{\odot 2}, \tag{A.19}$$

$$\ell_{bc} = \log\big(\max(v_{bc}, \varepsilon)\big), \tag{A.20}$$

$$s_{bc} = \mathrm{LayerNorm}(\mu_{bc} \| \ell_{bc}). \tag{A.21}$$

The maximum and logarithm are element-wise, with $\varepsilon = 10^{-8}$. Each of the eight heads has feature width $d_h = D/A$ and computes

$$q_{bhi} = \mathrm{LayerNorm}(W_h^Q s_{bi}), \tag{A.22}$$

$$k_{bhj} = \mathrm{LayerNorm}(W_h^K s_{bj}). \tag{A.23}$$

The spatial weights are

$$e_{bhij} = q_{bhi}^\top k_{bhj}/\sqrt{d_h} + b_{hij}, \tag{A.24}$$

$$A_{bhij} = \frac{m_{bj}\exp(e_{bhij})}{\sum_r m_{br}\exp(e_{bhir})}. \tag{A.25}$$

Values retain the temporal axis: $v_{bhjn} = W_h^V U_{bjn}$. For each time index, the attention output is

$$S_{bin}^{\mathrm{attn}} = W_O \left(\Big\|_{h=1}^{A} \sum_j A_{bhij} v_{bhjn} \right). \tag{A.26}$$

A single spatial weight matrix per window and head is therefore reused across all temporal tokens. The spatial residual branch receives S^{conv} in stages one and two and $S^{\mathrm{conv}} + S^{\mathrm{attn}}$ in stages three and four. The geometric bias informs attention about relative electrode locations; the channel embeddings and convolutional mixing nevertheless retain a fixed channel indexing, so this construction does not imply arbitrary-montage or channel-permutation invariance.

Bidirectional temporal state-space processing. Temporal dependencies are modeled with two independently parameterized Mamba2 modules [5]. After an internal RMSNorm, the tensor is reshaped to (BC, N, D), treating each electrode as a separate sequence. Writing R for reversal along the token axis and $\bar{U} = \mathrm{RMSNorm}(U)$, the bidirectional output is

$$F_{\mathrm{temp}}(U) = \mathcal{M}_{\rightarrow}(\bar{U}) + R\big[\mathcal{M}_{\leftarrow}(R[\bar{U}])\big]. \tag{A.27}$$

Reversing the backward output restores the original token alignment before summation. Each Mamba2 module uses state dimension 128, expansion factor two, local convolution width four, and head dimension 64. This module mixes information across time within each electrode; interactions between electrodes are provided by the spatial branch.

For completeness, the selective state-space core of a single Mamba2 head can be expressed in recurrent form as

$$S_n = \eta_n S_{n-1} + b_n v_n^\top, \tag{A.28}$$

$$o_n = S_n^\top c_n + d\, v_n, \tag{A.29}$$

where $S_n \in \mathbb{R}^{128\times 64}$, $v_n \in \mathbb{R}^{64}$, and $b_n, c_n \in \mathbb{R}^{128}$. The scalar transition η_n and the input and readout vectors depend on the input; b_n includes the input scaling from discretization, and d is a learned direct-path coefficient. This scalar-identity transition is the structured state-space component of Mamba2 [5]. The complete module also includes its input projections, local convolution, gating, normalization, and output projection. Equation (A.27) combines two such complete modules in opposite directions.

Local temporal convolution and gated feed-forward mapping. The Conformer branch supplements the state-space module with a local convolution. A linear map expands U from D to $4D$ features and splits it into $a, b \in \mathbb{R}^{2D}$. The gated features are $v = a \odot \sigma(b)$, where σ is the sigmoid. A depthwise temporal convolution of kernel size 15, with padding that preserves N, is followed by single-group GroupNorm, SiLU, and a projection back to D:

$$F_{\text{conv}}(U) = W_2\, \mathrm{SiLU}(\mathrm{GN}(\mathrm{DWConv}_{15}(v)))\,. \tag{A.30}$$

Finally, the feature-wise SwiGLU branch is

$$F_{\text{ffn}}(U) = W_3\big(\mathrm{SiLU}(W_1 U) \odot W_2 U\big). \tag{A.31}$$

Its hidden width is $\lfloor 8D/3 \rfloor$ rounded upward to the nearest multiple of 64. All four operations are wrapped by the residual update in Equation (A.12).

A.3 Training Pipeline Overview

Multiview joint-embedding learning. LeJEPA combines a joint-embedding agreement objective with Sketched Isotropic Gaussian Regularization [1], abbreviated SIGReg. Our EEG adaptation applies both terms directly to the global readout in Equation (A.11). It uses a single shared encoder, with no separate teacher, stop-gradient target, predictor network, or additional projection head. The training objective does not reconstruct EEG signals.

For each source window X_b, we sample $V_g = 2$ global views and $V_\ell = 6$ local views. We denote their union by $x_b^{(v)} = \mathcal{T}_v(X_b)$, $v = 1, \ldots, V$, where $V = V_g + V_\ell = 8$ and the first V_g indices designate the global views. The candidate augmentations include additive noise, global and channel-wise amplitude scaling, temporal shifts, and temporal, frequency, and spatial masking. These transformations have not been validated as preserving physiological markers or removing artifacts. The same presence codes a_b accompany all views of a window; spatial masking is an augmentation of observed signals rather than a change to the recorded channel-availability metadata. Each view is passed through the same model:

$$y_b^{(v)} = g_\theta(x_b^{(v)}, a_b; P) \in \mathbb{R}^{D_z}. \tag{A.32}$$

Here g_θ denotes the global component of f_θ. Stacking these outputs yields tensors of shape $B \times V_g \times D_z$ and $B \times V_\ell \times D_z$, as shown in Figure 7.

Alignment with the global-view center. The target for every view of window b is the mean of its global embeddings,

$$c_b = \frac{1}{V_g} \sum_{v=1}^{V_g} y_b^{(v)}. \tag{A.33}$$

Both global and local embeddings contribute to the alignment loss,

$$\mathcal{L}_{\text{align}} = \frac{1}{BVD_z} \sum_{b=1}^{B} \sum_{v=1}^{V} \|y_b^{(v)} - c_b\|_2^2. \tag{A.34}$$

The factor D_z^{-1} normalizes the loss by the feature dimension. Gradients propagate through both the individual embeddings and the center c_b. Alignment alone admits a constant representation, motivating the additional distributional constraint.

Sketched isotropic Gaussian regularization. At each optimization step, we sample $K = 256$ unit directions $u_k = \omega_k / \|\omega_k\|_2$, with $\omega_k \sim \mathcal{N}(0, I_{D_z})$. For a fixed view v, the one-dimensional projections $s_{bvk} = u_k^\top y_b^{(v)}$ define an empirical characteristic function

$$\widehat{\varphi}_{vk}(t) = \frac{1}{B} \sum_{b=1}^{B} \exp(\mathrm{i} t s_{bvk}). \tag{A.35}$$

The target is the characteristic function of a standard normal variable, $\varphi_0(t) = \exp(-t^2/2)$. Define

$$e_{vk}(t) = |\widehat{\varphi}_{vk}(t) - \varphi_0(t)|^2 \varphi_0(t). \tag{A.36}$$

The per-view regularizer approximates

$$\mathcal{R}_v = \frac{B}{K} \sum_{k=1}^{K} \int_{-5}^{5} e_{vk}(t)\, \mathrm{d}t \tag{A.37}$$

using the trapezoidal rule on $J = 17$ equally spaced points $t_j = -5 + 10j/(J-1)$, $j = 0, \ldots, J-1$:

$$\mathcal{R}_v \approx \frac{B\Delta t}{2K} \sum_{k=1}^{K} \sum_{j=0}^{J-2} \big[e_{vk}(t_j) + e_{vk}(t_{j+1})\big], \tag{A.38}$$

where $\Delta t = 10/(J-1)$. The complex discrepancy is evaluated through its real and imaginary parts,

$$|\widehat{\varphi} - \varphi_0|^2 = (\Re\widehat{\varphi} - \varphi_0)^2 + (\Im\widehat{\varphi})^2. \tag{A.39}$$

This operates on the unnormalized embeddings, so the unit-variance target also constrains their scale. The final regularizer is

$$\mathcal{L}_{\text{SIG}} = \frac{1}{V} \sum_{v=1}^{V} \mathcal{R}_v. \tag{A.40}$$

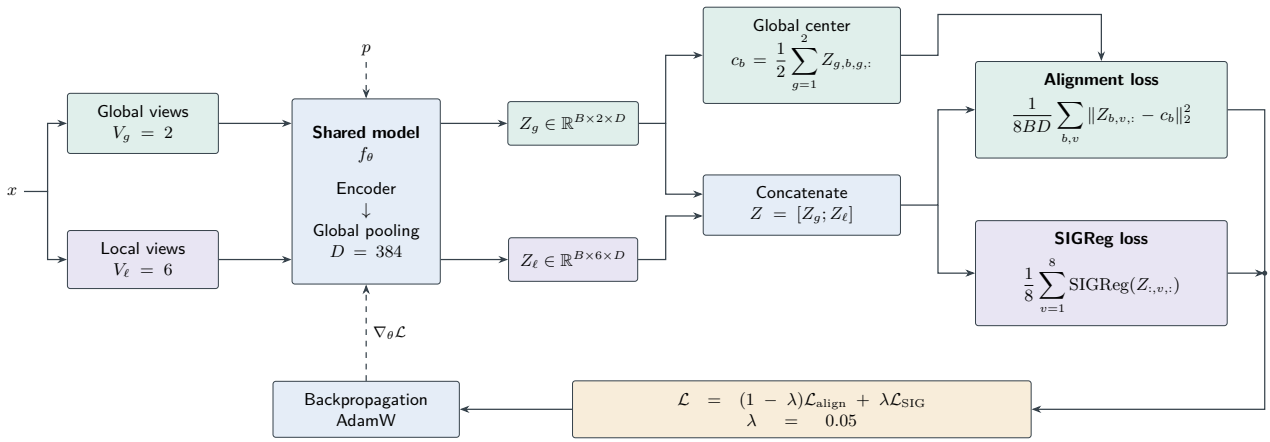


Figure 7: Multiview LeJEPA training. Two global and six local views are encoded by the same shared model. All view embeddings are aligned with the mean of the global embeddings of their source window. SIGReg is evaluated across the batch separately for each view and then averaged. The weighted loss updates the complete encoder and global readout. The global center remains differentiable.

Thus, views are not pooled together as independent samples for SIGReg. The same random directions are reused across views within a step by seeding the sampler with the global step index. In distributed training, characteristic-function sums and sample counts are aggregated across workers; B in Equations (A.35), (A.37) and (A.38) then denotes the aggregated batch size. The regularizer is evaluated in float32 when embeddings are produced in reduced precision.

Objective and optimization. The complete objective is

$$\mathcal{L} = (1-\lambda)\mathcal{L}_{\text{align}} + \lambda\mathcal{L}_{\text{SIG}}, \qquad \lambda = 0.05. \tag{A.41}$$

The proposed training uses normalized TUEG windows with mapped channels, excluding the TUAB and TUAR subsets. The optimizer is AdamW with initial learning rate 10^{-4}, momentum coefficients $(0.9, 0.95)$, and weight decay 0.05. Biases, normalization parameters, embeddings, learned queries, LayerScale vectors, and the spatial bias table are excluded from weight decay. The total gradient norm is clipped at one. A cosine learning-rate schedule is stepped after every optimizer update, with a 10,000-step horizon and minimum learning rate 10^{-5}. The prescribed schedule uses batches of 16, bfloat16 mixed precision, and gradient checkpointing within encoder blocks. Its duration is 20 epochs, capped at 500 training batches and 100 validation batches per epoch.

The objective encourages agreement under the selected view transformations while penalizing departures from the Gaussian projection target. Neither term explicitly removes subject identity, acquisition-related information, or artifacts shared across views. Whether the resulting representations retain useful information under montage changes must therefore be established by downstream evaluations and ablations.